

REST API Rules – Complete Guide

1. What is REST

REST stands for Representational State Transfer. It is a set of architectural rules for designing APIs that communicate between two systems using HTTP in a standardized way.

2. Resource-Based URLs

REST APIs must be resource-oriented. URLs should represent resources (nouns), not actions (verbs).

Correct: /users, /events/10

Incorrect: /getUsers, /deleteEvent

3. Use HTTP Methods Correctly

HTTP methods define the action:

GET – Read data

POST – Create data

PUT – Update full data

PATCH – Update partial data

DELETE – Remove data

4. Path Variables for Identity

Path variables are used to identify a specific resource.

Example: GET /users/10

5. Query Parameters for Filtering & Sorting

Query parameters are used for searching, filtering, and sorting.

Example: GET /events?status=ACTIVE&type=RFP

6. Request Body for State Representation

Request body is used to send the representation of a resource when creating or updating it.

Example: POST /users with JSON body

7. Stateless Communication

Each request must be independent. The server should not store client state.

Authentication tokens must be sent with every request.

8. Representation of Data

REST APIs transfer representations of resource state (usually JSON or XML), not database entities.

9. Proper HTTP Status Codes

REST APIs must use standard HTTP status codes.

200 OK, 201 Created, 204 No Content, 400 Bad Request, 401 Unauthorized, 404 Not Found, 500 Server Error

10. Consistent Naming & Plural Resources

Use plural nouns for resources.

Correct: /users, /orders

Incorrect: /user, /order

11. Versioning

APIs should be versioned to support backward compatibility.

Example: /api/v1/users

12. Golden REST Rule

URL describes WHAT the resource is.

HTTP method describes WHAT action to perform.